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ABSTRACT— The proliferation of e-commerce is apparent as retailers actively participate in digital interactions with customers. Users seek a good user experience coupled with exceptional products and services. This study focused on the enhancement of the user experience on electronic commerce websites. A two-method usability assessment strategy was employed to examine the impact of usability components on the user experience, both in terms of facilitation and hindrance. This method encompassed doing usability testing with end-users and expertly evaluating heuristics. The usability testing uncovered interface deficiencies and purchasing preferences influenced by the content and user journey. The heuristic evaluation primarily concentrated on assessing the user interface and interaction elements. Both techniques discovered numerous usability problems. This study proposes design principles for e-commerce websites that prioritize the user experience.

Keywords—: E-commerce usability; usability evaluation; user testing; heuristic evaluation; user experience.

I. INTRODUCTION

Companies are increasingly diversifying their range of commercial services through online platforms. To maintain a competitive edge, it is essential for businesses to go beyond simply giving high-quality presentations on their websites. According to Vaataja et al. [1], it is equally important for them to place a priority on the creation of a user experience that is suitable for themselves. The user interface of a website has practically taken the position of the sales advisor. The purpose of this interface is to provide clients with relevant information and functionalities that are essential to their success, therefore making it easier for them to achieve their goals. The user interface of an online business has the capacity to either positively influence clients to make a purchase or adversely prevent them from making a purchase, which ultimately results in a decrease in the financial performance of the firm [2]. The effectiveness of electronic commerce is consequently influenced by the user's experience with the website [3]. It is crucial for stakeholders to be aware of the concept of user experience in all aspects of the business environment to ensure that the user have a positive experience. The design of websites is becoming an increasingly important factor for businesses that want to maximise their profits in a market that is extraordinarily competitive [4]. Clemmensen et al. [5] postulate that interface heuristics and web standards are often portrayed as concepts

that can be applied generally rather than being tailored to specific businesses or target markets. However, it is essential to keep in mind that every sector has its own unique objectives and targets, which is why it is necessary to establish guidelines to make it easier to achieve these goals. Furthermore, Barnard and Wesson [6] state that despite the proliferation of multiple standards for web design, extensive international research has demonstrated that a significant majority of websites continue to exhibit poor usability design. The utilisation of these guidelines in practical applications lacks substantial evidence. According to Väänänen-Vainio-Mattila and Wäljas [7], the predominant emphasis of interface heuristics and standards lies in usability rather than the more contemporary notion of user experience. The objective of this study is to pinpoint the precise factors of usability that either increase or impede users' performance and to establish design principles that will assist in the development and enhancement of e-commerce websites.

A. Research Question

How can a dual-method usability assessment of e-commerce platforms enhance the overall user experience. The primary research inquiries have resulted in the further sub-inquiries listed below:

- Which components of usability enhance the user experience offered by an e-commerce website?
- What techniques can be applied to measure the usability and user experience of an e-commerce website?
- What are the design standards that can optimize the design and development of e-commerce platforms?

B. Value of the Study

E-commerce website developers will benefit most from this study. Website designers can use the study to determine which usability variables improve the user experience and build accordingly. The findings of the research could improve existing websites. This study will aid retail sectors because many e-commerce rules are unsupported by empirical data, making them unsuitable for certain situations Lowry et al. [8]. Furthermore, the impact of user experience in e-commerce is understudied; thus, this study is significant Vainio-Mattila and Wäljas [7]. Users will benefit from the study by having access to user-friendly and enjoyable websites.

II. LITERATURE REVIEW

Technology has made website design and visual display crucial for companies trying to maximise earnings in a competitive market. This is because more companies are going online. Lee and Koubek [4]. Within the domain of internet commerce, the user assumes a crucial role in attaining enhanced financial profits. Hence, it is imperative for the website designer to establish a robust connection with the users by comprehending their decision-making process and the desired path they wish to pursue. Users subjectively choose one product over another based on quality or desirability. Researchers describe preference as a user's subjective evaluation of a website's interface. This assessment affects their website usage and decision-making. Lee and Koubek [4]. According to McCall's [9], consumers who conduct online research before visiting physical outlets make 40% to 50% of offline retail purchases. More than 70% of internet buyers say a company's substandard website adversely impacts their impression of it. The Internet is a crucial platform for brands to communicate with customers, promoting efficient communication and building strong relationships.

A. Usability

Sharp et al. [10] define usability as the quality of an interactive product being easy to learn, effective to use, easy to remember, and pleasurable from the user's standpoint. Human-computer interaction (HCI) research on e-commerce website efficacy has mostly focused on website usability. Petre et al. [11].

Nielsen [12] wrote in "Usability Engineering" that usability is not a single, one-dimensional property of the user interface. Usability is judged on five factors: learnability, efficiency, memorability, mistakes, and satisfaction. The lack of usability elements is a significant factor contributing to the failure of numerous e-commerce websites to attract clients [13]. Clients who are unable to locate the desired item on an e-commerce website are less likely to engage in online shopping. Head [14] suggested that an optimal website interface should possess characteristics such as ease of learning, ease of memorization, user friendliness, and the ability to facilitate error recovery. The usability of a product is not an attribute of the product alone. It is a characteristic of how users engage with a product within a certain situation of use [15].

The usability of a product can vary significantly depending on the environment in which it is used. Therefore, it is important to clearly specify the context for design and assessments, as stated in ISO 9241-11. The context encompasses the individuals utilising the product, their objectives, activities, and tools, as well as the physical and social environment in which the product is used. Culture is a crucial factor to consider when designing e-commerce platforms, in addition to trust and usability. Nante and Glaser [16] study examined the influence of language and culture on the perceived usability of websites. They discovered that when websites are developed in the user's native language, the perceived usability is enhanced.

According to Van Iwaarden et al. [17], language, culture, and religion can significantly influence a user's perception of a website. However, Clemmensen et al. [5] argued that cultural concerns have a minimal impact on most research regarding usability evaluation methods. The cultural background of users is seldom documented; work scenarios are thought to be

culturally impartial; interface heuristics and Web standards are portrayed as universal and not specified for specific industries or target markets.

Learnability, memorability, mistake recovery, efficiency, and satisfaction are the five main factors that determine a system's usability overall. These components provide irrefutable evidence of a system's usefulness. This evidence is prevalent in the corpus of published research.

However, to enhance the system's usability and acceptance among users, it is important to consider factors such as culture, language, and religion throughout the system's development. It is crucial to evaluate these components in the current period of heightened competition in e-commerce.

B. User Experience

In the past, conventional human-computer interaction (HCI) has primarily assessed the usability of a system based on criteria such as efficiency, utility, effectiveness, safety, learnability, and memorability. However, it has neglected to consider the significance of additional dimensions of the user experience, including aesthetic appeal and enjoyment. Bevan [18] argued that the user experience is influenced by several essential characteristics, including learnability, usability, usefulness, and aesthetic appeal. The user experience elements pertain to subjective attributes and mostly revolve around the feelings that users have towards a system. These aspects are distinct from the more objective usability aims, as they focus on how users perceive and interact with the product from their own standpoint [10].

According to Hassenzahl [19], the perspective of user experience (UX) expands upon the user-centred design approach by addressing concerns that go beyond actual functioning and usability. He discussed how the field of user experience (UX) encompasses various aspects of users' motives and emotions, including stimulation, identification, and self-expression. These expressions may encompass both positive and negative sentiments. The first choice encompasses users' perceptions of a system or site as being fulfilling, enjoyable, engaging, entertaining, helpful, motivating, visually pleasing, and rewarding. The latter scenario encompasses a range of emotions, such as frustrating, annoying, and boring Sharp et al. [10].

The individual's subjective emotions can impact their future interactions with the system and may also be transmitted to other users, potentially influencing their subjective experience as well [20]. Despite the development of numerous user experience (UX) models in the past decade, there has been a lack of systematic study on the specific features that impact the UX of web services Väänänen-Vainio-Mattila and Wäljas [7]. Sward [21] defines user experience as the end user's interaction with a product, service, and organisation. The concept of user experience pertains to the inherent value that is obtained through a user's engagements with a particular product or service, considering the contextual factors such as time, location, and the user's disposition. The value that users derive from a product or service can be categorized as either actual value, which encompasses factors such as efficiency and effectiveness, or perceived value, which includes elements such as trustworthiness, emotions, satisfaction, aesthetic appeal, and entertainment. In some cases, the value experienced by users may be a combination of both actual and perceived factors.

According to Rubinoff [22], the concept of user experience (UX) encompasses four distinct variables, including

branding, usability, functionality, and content. Rubinoff asserts that while these qualities in isolation may not contribute to a favourable user experience, their combination can serve as the fundamental elements for achieving success in website design.

The diagram depicted in Figure 1 illustrates a conceptual framework wherein user experience is portrayed as a comprehensive word encompassing several components that contribute to the overall user experience.

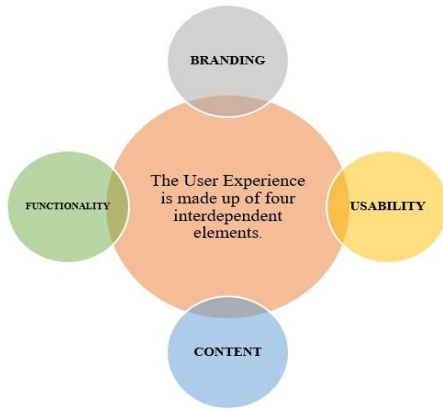


Figure 1: Four elements of user experience [22]

To ensure a satisfactory user experience, it is necessary to incorporate a diverse array of user experience (UX) disciplines.

Multiple studies have conclusively shown that the user experience extends beyond system functionality. Instead, it is predominantly influenced by the user's emotions, which can be interpreted as their overall perception of the product.

Based on the previous research, it is crucial to include a user-centric approach in the design of products to ensure their widespread acceptance among consumers.

C. User Experience Methodologies

Usability engineering focuses on cognitive or behavioural user experiences. However, a modern trend is to explore the emotional experiences of users by representing their reactions to designs through verbal and visual methods. Specialised user experience heuristics are also being created. There are several toolkits for typical usability evaluations, but user experience evaluation is different. Stopwatches and logging are unsuitable for assessing the user experience, which is subjective. Law et al. [23]. However, assessors must determine the user's subjective system perception. User motivation and expectations influence the experience more than usability Mäkelä and Fulton [24]. Some of the techniques used in measuring user experience include:

1) Face Reader

This device can analyse facial expressions in real-time. You can always tell someone is likely to be experiencing one of the six fundamental emotions—joy, anger, sadness, surprise, fear, or disgust—just by looking at their face Bernhaupt et al. [25].

2) Prolonged diary method

A product prototype is given to users for 6–12 months. Users record their thoughts and feelings in diary entries at set intervals Bolger et al. [26].

D. Usability Evaluation Methods

Several usability evaluation methods have been devised to detect usability issues. Nielsen and Mack [27] categorised usability evaluation methods into four broad groups: automatic which employs software to assess a user interface, empirical which involves real users interacting with a user interface, formal which utilizes models to evaluate a user interface, and informal where evaluators rely on rules as well as their expertise, knowledge, and experience to evaluate an interface. In contrast, Grey and Salzman [28] employed a binary classification system to categorize these methods: analytic and empirical. The analytic side encompasses techniques such as heuristic evaluation and cognitive walkthrough, whereas empirical techniques involve methodologies and procedures known as user testing.

III. CONCLUSION

The purpose of this research was to define usability and user experience and to investigate the usability evaluation methodologies that can be used to evaluate the usability and user experience of e-commerce websites. Usability testing and heuristic evaluation were both employed in the study's research strategy. The two approaches were evaluated for their suitability, efficiency, and satisfaction in assessing the usability of online shopping websites. They worked together in a complementary and validating manner, creating a synergistic effect. The best way to evaluate e-commerce websites is with a combination of heuristic evaluation and user testing. There was a strong correlation between the usability of a website and the satisfaction of its users, according to the findings. This study found that website features marked by inadequate usability significantly impede users' overall experience. When encountering uneven navigation structures, product-based navigation menus, difficult-to-understand functionality, and inadequate presentation of information offers. Conversely, optimal usability improves the quality of the whole user experience.

Extensive guidelines for improving the e-commerce platform user experience were developed in large part from the results of the user testing and heuristic evaluation. The research findings suggest that designers and developers should employ appropriate user interface strategies, vital e-commerce attributes, and, most importantly, user expectations. The following design principles must be adhered to by web designers after understanding the customer's needs:

Map out the user's desired purchasing process to construct the user journey.

Craft and organise content to meet user requirements with particular emphasis on the presentation, such as ensuring that decision-making information is prominently displayed within the architecture.

The website design should be creative and compelling to keep users coming back.

In conclusion, it is imperative that the design and development of e-commerce sites prioritise the needs of users as the primary motivating factor.

A. Further Research

- Research should be conducted to explore the connections between usability and user experience.
- The guidelines provided should be used in developing novel websites.
- Research should be conducted to investigate the influence of culture on user experience and usability of e-commerce website.

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